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Stability and instability of geodesic flows on smoothly deformed $\mathcal{S}^2$

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ABSTRACT

Starting from a two-dimensional smoothly perturbed Riemannian manifold, a sphere $\mathcal{S}^2$, we have explored how the properties of stability and instability of the geodesic flows are modified under geometrical perturbations of the metric tensor, we observed some memory for the metric tensor structure of the geodesic flows for small perturbations (weak chaoticity), which totally disappeared for large perturbations (strong chaoticity). We quantified this transition from weak to strong chaos by computing the largest Lyapunov exponent along with solving the Jacobi-Levi-Civita equation, a similar treatment have been started for a Torus $\mathcal{T}^2$.

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1 Internship overview

Throughout this research internship, even from the beginning, thanks to Pr.Pettini, the goal was pretty clear and well posed : it was firstly to build a simple toy model to visualize the common idea of chaos (whether weak or strong), then to link it to some quantitative quantities and observe directly their behavior according to the degrees of chaoticity of the system. We propose in this first section to summarize in a simpler manner the different steps we were having to pass through all along this work.

All the numerical computations have been worked from scratch using Python language and related packages: numpy, sympy and matplotlib among others, it consists largely of symbolic calculus and resolution of system of differential equations (Ode's), 3d plots have been realized on Wolfram Mathematica 13.0 with the help of Dr.Matteo Gori.

- Firstly, we had to lay down the theoretical framework, notions such (*weak & strong*) *chaos* at the heart of our studies, but also some peripheral notions such the *Integrability* and *geodesics flows*, then we prepared the ground by introducing the notion of *Largest Lyapunov exponent* along with the *Jacobi-Levi-Civita*

equation which will later serve as a probe, all that have been resumed in [section 2](#).

- We was then having to choose the simplest model physicist like to use : A *Sphere* metric. We recapitulated in [subsection 3.1](#) the basic notions of it, then solving the geodesic equations numerically using Runge-Kutta ($\mathcal{O}^4$).
- Then we settled the general form of the perturbed metric as explained in [subsection 3.2](#), picked a specific perturbation and resolve the geodesic equations again for different values of parameters (ϵ, κ) and understand carefully the behavior of those geodesics.
- After theses visual achievements, we turned to quantitative studies, based on theoretical explanations given previously in [section 2](#), we builded the corresponding discretized version and discussed the different shapes we obtain in [section 4](#) and [section 5](#).
- Further discussions, important conclusions and other directions are available in [section 6](#), a similar work have been started for the metric of a Torus $\mathcal{T}^2$ but not discussed here.

This work was generously supervised by Pr.Pettini from Aix-Marseille Univ, Université de Toulon, and Centre de Physique Théorique de Marseille : CPT. Please, feel free to share with us any constructive comments and valuable remarks when reading this report.

2 Geometrization of Hamiltonian Dynamics

The beginning of mathematical physics in mid-XVII century was successful to an incommensurate point, and is no longer to be discussed. Classical Newtonian dynamics showed us all his powers along decades, especially with the advent of Lagrangian and Hamiltonian dynamics, where the richness of certain theoretical models shown a particular behavior and needs a more careful treatment.

Before going further into the details, let's recap that we define a dynamical system as a system of equations which changes over time (i.e. equations of motion), which generally consist of a system of continuous differential equations (Ode's) which will be called a *flows* that have to be solved, theses dynamical systems models real world phenomena such as disease spread, flow of water in a pipe and even a "simple" Lorenz oscillator. Solving theses differential equations will give you the behavior of the whole system along time, a *Trajectory*, The trajectory of a flow is the path the flow takes as time progresses.

One characteristic of some dynamical systems is their sensitive dependence on initial conditions, also known as *Chaos*, chaos is a branch of mathematics by itself, and many definitions can be given, the one we'd like to use is,

Tiny differences in the initial conditions of the system will lead to large differences in the solutions.

A popular example is the so-called *Butterfly Effect*, but a natural question that arise immediately is : *how to differentiate between a chaotic and a non-chaotic dynamical system ?*

In [1, 2, 3, 4, 5] and references therein, it is explained that by identifying Hamiltonian flows with geodesic flows of suitably chosen Riemannian manifolds, we will be able to explain the origin of chaos in classical Newtonian dynamics and to quantify its strength. The arguments goes like that,

1. Lagrangian and Hamiltonian formulations are known to be the variational formulation of dynamics through Mauritius' least action principle.
2. The geodesics of a Riemannian manifold fulfill a variational principle also !
3. Thus, the possibility of rephrasing classical dynamics in geometrical terms by identifying a Hamiltonian flow with a geodesic flow on a Riemannian manifold that will provides a real practical resource for studying the stability/instability properties of Hamiltonian dynamics : Chaos.

In fact, the geometrization of dynamics allows us to borrow from geometry two important notions commonly-known as the *Lyapunov exponent* and *Jacobi-Levi-Civita (JLC) equation*.

Lyapunov exponent

The Lyapunov exponents λ is a tool that is very useful for determining whether the system is strongly dependant on initial conditions or not (i.e. chaotic). It tells us whether or not two points in the phase space of a dynamical system, that are initially very close to each other, stay close in the subsequent motion. In other words, they measure the average rate of divergence or convergence of nearby orbits. The exponential divergence of orbits, in a practical sense, implies the lost of predictability of the system, a system with positive Lyapunov exponent is defined as chaotic.

For small initial distances δ_0 and short times t . The distance between the two trajectories as a function of time is then given by,

$$\delta(t) = \delta_0 e^{\lambda t} \quad \rightarrow \quad \lambda = \frac{1}{t} \ln \frac{\delta(t)}{\delta_0} \quad (2.1)$$

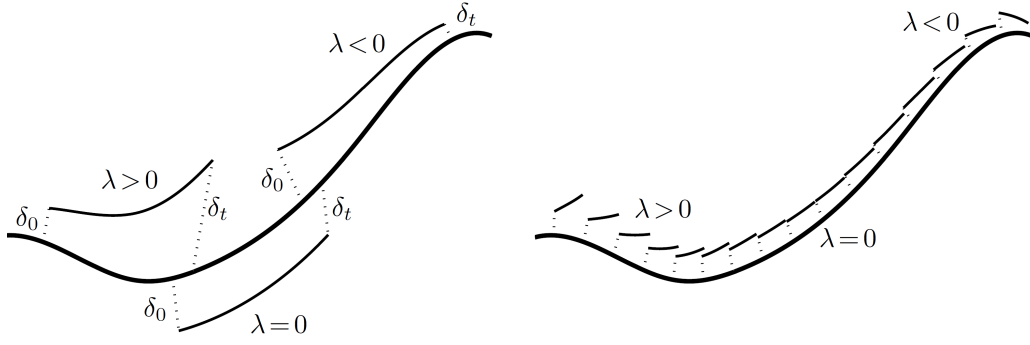


Figure 1. Left: Diverging, parallel and converging trajectories correspond to a local divergence rate of $\lambda > 0$, $\lambda = 0$ and $\lambda < 0$, respectively. Right: A Lyapunov exponent is determined as an average of the local divergence rate. Taken from [6].

where for $\lambda > 0$ the trajectories are diverging, for $\lambda < 0$ they are converging and for $\lambda = 0$ the distance between corresponding points on the two trajectories does not change. All of them are illustrated in Figure 1.

Jacobi-Levi-Civita equation

In simple words, the Jacobi-Levi-Civita (JLC) equation links between the curvature features of a given manifold and the stability/instability properties of its geodesics and it describe the evolution of the so-called geodesic separation vector field.

In Riemannian geometry, a Jacobi field J is a vector field along a geodesic γ , where J locally measures the distance between nearby geodesics (i.e. describing the difference between the geodesic and an "infinitesimally close" geodesic) which in a parallel-transported frame reads,

$$\frac{d^2 J^k}{ds^2} + R^k_{ijr} \frac{dq^i}{ds} J^j \frac{dq^r}{ds} = 0. \quad (2.2)$$

where R^k_{ijr} are the components of the Riemann curvature tensor, since the JLC equation involves the Riemann curvature tensor, the relevant geometric properties must be related to curvature. The holy grail of Riemannian geometrization of the dynamics is an equation relating (in)stability with geometry on some global information about the average degree of instability (chaos) of the dynamics. Interested readers may consult [2, 3, 7] for more details. In summary, here is the idea,

- If the solution of Equation 2.2 computed along a given trajectory remains bounded or grows at most linearly with time, then this trajectory is stable.
- If it grows exponentially with time, then the trajectory is chaotic, (i. e. parametrically unstable).

3 2-Sphere manifold

In this sections, we introduce the systems of equations for the toy model builded in this work, it consist of a system of two geodesic equations (θ, ϕ) each time for two free parameters (ϵ, κ) .

3.1 2-sphere recap'

In terms of Cartesian coordinates, the surface of a sphere can be described as,

$$\begin{aligned} x &= r \cos(\theta) \sin(\phi) \\ y &= r \sin(\theta) \sin(\phi) \\ z &= r \cos(\phi) \end{aligned} \tag{3.1}$$

with fixed radius r , θ and ϕ are the angular spherical coordinates such that $0 \leq \theta < 2\pi$ and $0 \leq \phi \leq \pi$, the metric in the (x, y, z) coordinates writes as,

$$ds^2 = dx^2 + dy^2 + dz^2 \tag{3.2}$$

Is it more convenient to use spherical coordinates (r, θ, ϕ) , a change of coordinates from X to Y is given by,

$$g_{ij} = \sum_{\alpha=1}^3 \left(\frac{\partial X^\alpha}{\partial Y^i} \right) \left(\frac{\partial X^\alpha}{\partial Y^j} \right) \tag{3.3}$$

Then as habitual, the metric of a 2-sphere is given by,

$$ds^2 = r^2 d\theta^2 + r^2 \sin^2(\theta) d\phi^2 \tag{3.4}$$

The metric [3.4](#) can also be expressed in a more convenient way as,

$$g_{ij} = \begin{pmatrix} r^2 & 0 \\ 0 & r^2 \sin^2(\theta) \end{pmatrix} \tag{3.5}$$

In order to obtain the "best" trajectory (i.e. the minimal distance between two point on a manifold), we have to solve the well-known geodesic equations,

$$\frac{d^2 x^i}{ds^2} + \Gamma^i_{jk} \frac{dx^j}{ds} \frac{dx^k}{ds} = 0 \tag{3.6}$$

where s is a scalar parameter and Γ^i_{jk} are the Christoffel symbols, known also as connections,

$$\Gamma^i_{jk} = \frac{1}{2}g^{il}(\partial_j g_{lk} + \partial_k g_{jl} - \partial_l g_{jk}) \quad (3.7)$$

From 3.5, the only non-vanishing connections are given by :

$$\begin{aligned} \Gamma^\theta_{\phi\phi} &= -\sin(\theta) \cos(\theta) \\ \Gamma^\phi_{\theta\phi} &= \frac{\cos(\theta)}{\sin(\theta)} \end{aligned}$$

To solve (numerically) the system of equations 3.6, we used the so-called Runge-Kutta method up to order 4, which gives results as shown in,

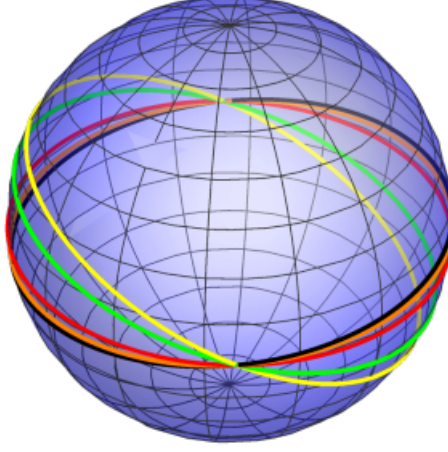


Figure 2. Some random unperturbed-sphere geodesics, this kind of trajectories will be labeled as trajectories of 1st kind.

As already known, the geodesics on a surface of a sphere are arc of circles, we done so-far to calibrate our code.

3.2 Perturbed 2-sphere

In order to achieve our goals, we have modified the metric in such a way that the radius become a function by itself as $r = r + R(\theta, \phi)$, accordingly and coming back to eqs.3.1 and 3.3, we obtain the generic form of the metric will as follows,

$$\tilde{g}_{ij} = \begin{pmatrix} (r + R(\theta, \phi))^2 + \left(\frac{\partial R(\theta, \phi)}{\partial \theta}\right)^2 & \frac{\partial R(\theta, \phi)}{\partial \theta} \frac{\partial R(\theta, \phi)}{\partial \phi} \\ \frac{\partial R(\theta, \phi)}{\partial \phi} \frac{\partial R(\theta, \phi)}{\partial \theta} & ((r + R(\theta, \phi)) \sin(\theta))^2 + \left(\frac{\partial R(\theta, \phi)}{\partial \phi}\right)^2 \end{pmatrix} \quad (3.8)$$

For next purposes, r will be fixed to unity. As we did earlier, we re-compute and re-resolve the geodesic equations 3.6 for the metric 3.8, to do so, we computed again the connections, (for shortness, R must to be understood as $R(\theta, \phi)$)

$$\Gamma_{\theta\theta}^{\theta} = \frac{\left(\left(R - \frac{\partial^2}{\partial\theta^2} R + 1 \right) \left(\frac{\partial}{\partial\phi} R \right)^2 + \left(R + \frac{\partial^2}{\partial\theta^2} R + 1 \right) \left(R^2 \sin^2(\theta) + 2R \sin^2(\theta) + \sin^2(\theta) + \left(\frac{\partial}{\partial\phi} R \right)^2 \right) \right) \frac{\partial}{\partial\theta} R}{\left(R^2 \sin^2(\theta) + 2R \sin^2(\theta) + \sin^2(\theta) + \left(\frac{\partial}{\partial\phi} R \right)^2 \right) \left(R^2 + 2R + \left(\frac{\partial}{\partial\theta} R \right)^2 + 1 \right) - \left(\frac{\partial}{\partial\phi} R \right)^2 \left(\frac{\partial}{\partial\theta} R \right)^2}$$

$$\Gamma_{\theta\phi}^{\theta} = \frac{\left(R \frac{\partial}{\partial\phi} R + \frac{\partial}{\partial\phi} R + \frac{\partial}{\partial\theta} R \frac{\partial^2}{\partial\theta\partial\phi} R \right) \left(R^2 \sin^2(\theta) + 2R \sin^2(\theta) + \sin^2(\theta) + \left(\frac{\partial}{\partial\phi} R \right)^2 \right) - \left(\frac{R^2 \sin(2\theta)}{2} + R \sin^2(\theta) \frac{\partial}{\partial\theta} R + R \sin(2\theta) + \sin^2(\theta) \frac{\partial}{\partial\theta} R + \frac{\sin(2\theta)}{2} + \frac{\partial}{\partial\phi} R \frac{\partial^2}{\partial\theta\partial\phi} R \right) \frac{\partial}{\partial\phi} R \frac{\partial}{\partial\theta} R}{\left(R^2 \sin^2(\theta) + 2R \sin^2(\theta) + \sin^2(\theta) + \left(\frac{\partial}{\partial\phi} R \right)^2 \right) \left(R^2 + 2R + \left(\frac{\partial}{\partial\theta} R \right)^2 + 1 \right) - \left(\frac{\partial}{\partial\phi} R \right)^2 \left(\frac{\partial}{\partial\theta} R \right)^2}$$

$$\Gamma_{\phi\phi}^{\theta} = - \frac{\left(R \sin^2(\theta) + \sin^2(\theta) + \frac{\partial^2}{\partial\phi^2} R \right) \left(\frac{\partial}{\partial\phi} R \right)^2 \frac{\partial}{\partial\theta} R + \left(R^2 \sin^2(\theta) + 2R \sin^2(\theta) + \sin^2(\theta) + \left(\frac{\partial}{\partial\phi} R \right)^2 \right) \left(\frac{R^2 \sin(2\theta)}{2} + R \sin^2(\theta) \frac{\partial}{\partial\theta} R + R \sin(2\theta) + \sin^2(\theta) \frac{\partial}{\partial\theta} R + \frac{\sin(2\theta)}{2} - \frac{\partial^2}{\partial\phi^2} R \frac{\partial}{\partial\theta} R \right)}{\left(R^2 \sin^2(\theta) + 2R \sin^2(\theta) + \sin^2(\theta) + \left(\frac{\partial}{\partial\phi} R \right)^2 \right) \left(R^2 + 2R + \left(\frac{\partial}{\partial\theta} R \right)^2 + 1 \right) - \left(\frac{\partial}{\partial\phi} R \right)^2 \left(\frac{\partial}{\partial\theta} R \right)^2}$$

$$\Gamma_{\theta\theta}^{\phi} = - \frac{\left(\left(R - \frac{\partial^2}{\partial\theta^2} R + 1 \right) \left(R^2 + 2R + \left(\frac{\partial}{\partial\theta} R \right)^2 + 1 \right) + \left(R + \frac{\partial^2}{\partial\theta^2} R + 1 \right) \left(\frac{\partial}{\partial\theta} R \right)^2 \right) \frac{\partial}{\partial\phi} R}{\left(R^2 \sin^2(\theta) + 2R \sin^2(\theta) + \sin^2(\theta) + \left(\frac{\partial}{\partial\phi} R \right)^2 \right) \left(R^2 + 2R + \left(\frac{\partial}{\partial\theta} R \right)^2 + 1 \right) - \left(\frac{\partial}{\partial\phi} R \right)^2 \left(\frac{\partial}{\partial\theta} R \right)^2}$$

$$\Gamma_{\theta\phi}^{\phi} = - \frac{\left(R \frac{\partial}{\partial\phi} R + \frac{\partial}{\partial\phi} R + \frac{\partial}{\partial\theta} R \frac{\partial^2}{\partial\theta\partial\phi} R \right) \frac{\partial}{\partial\phi} R \frac{\partial}{\partial\theta} R + \left(R^2 + 2R + \left(\frac{\partial}{\partial\theta} R \right)^2 + 1 \right) \left(\frac{R^2 \sin(2\theta)}{2} + R \sin^2(\theta) \frac{\partial}{\partial\theta} R + R \sin(2\theta) + \sin^2(\theta) \frac{\partial}{\partial\theta} R + \frac{\sin(2\theta)}{2} + \frac{\partial}{\partial\phi} R \frac{\partial^2}{\partial\theta\partial\phi} R \right)}{\left(R^2 \sin^2(\theta) + 2R \sin^2(\theta) + \sin^2(\theta) + \left(\frac{\partial}{\partial\phi} R \right)^2 \right) \left(R^2 + 2R + \left(\frac{\partial}{\partial\theta} R \right)^2 + 1 \right) - \left(\frac{\partial}{\partial\phi} R \right)^2 \left(\frac{\partial}{\partial\theta} R \right)^2}$$

$$\Gamma_{\phi\phi}^{\phi} = \frac{\left(R^2 \sin^2(\theta) + 2R \sin^2(\theta) + \frac{R \sin(2\theta)}{2} \frac{\partial}{\partial\theta} R + R \frac{\partial^2}{\partial\phi^2} R + 2 \sin^2(\theta) \left(\frac{\partial}{\partial\theta} R \right)^2 + \sin^2(\theta) + \frac{\sin(2\theta)}{2} \frac{\partial}{\partial\theta} R + \frac{\partial^2}{\partial\phi^2} R \right) \frac{\partial}{\partial\phi} R}{R^3(\theta, \phi) \sin^2(\theta) + 3R^2 \sin^2(\theta) + R \sin^2(\theta) \left(\frac{\partial}{\partial\theta} R \right)^2 + 3R \sin^2(\theta) + R \left(\frac{\partial}{\partial\phi} R \right)^2 + \sin^2(\theta) \left(\frac{\partial}{\partial\theta} R \right)^2 + \sin^2(\theta) + \left(\frac{\partial}{\partial\phi} R \right)^2}$$

For our work, we choose a perturbation such that:

$$R(\theta, \phi) = \epsilon \cos(\kappa(\theta + \phi)) \quad (3.9)$$

The above geodesic equations 3.6 have been solved for the metric 3.8 and some results are shown in Figure 3. An important remark is that even for small ϵ , we may find trajectories of 3rd kind, especially for non-equatorial directions.

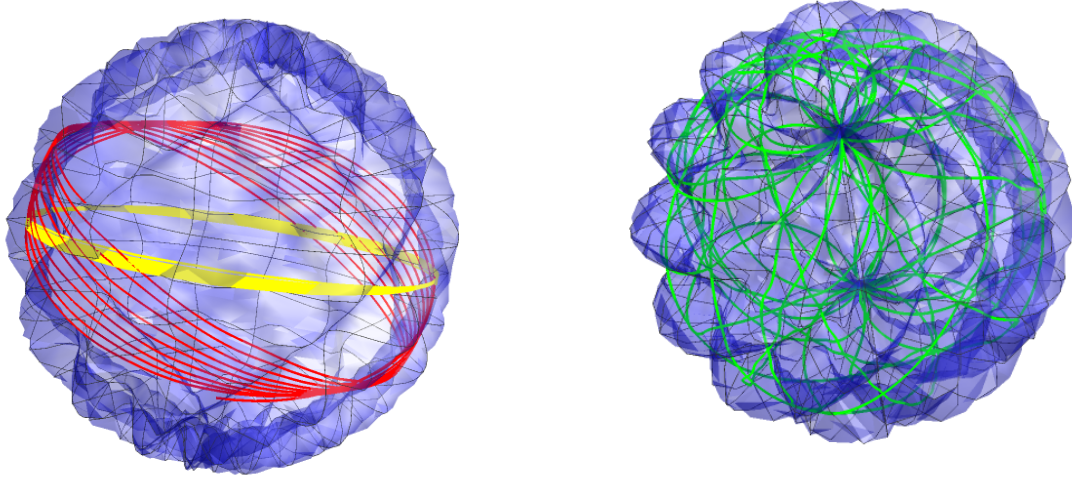


Figure 3. Geodesics of perturbed sphere with (ϵ, κ) parameters. Left: Trajectories of 2^{nd} kind for $(0.02, 15)$. Right: Trajectory of 3^{rd} kind for $(0.15, 10)$.

4 Largest Lyapunov exponent

After all the beautiful orbit-trajectories shown above of 1^{st} , 2^{nd} and 3^{rd} kind, it would be nice to have a simple measure that could discriminate among the types of orbits in the same manner as the we did visually. As previously explained in [section 2](#), the *Largest Lyapunov exponent* (based on orbit separation) could help us on that. We will focus on this sections to explain the used algorithm and discuss the obtained results.

The general idea is to follow two nearby orbits and to calculate their average logarithmic rate of separation, at each iteration, one of the orbits has to be moved back to the vicinity of the other along the line of separation, the complete procedure is as follows,

1. **Create a reference orbit:** Start with any initial condition $(x_0, y_0, z_0) \equiv (\theta_0, \phi_0)$, solve the geodesic equations for a certain time, here we solved up to 10^3 with a time step $\Delta t = 0.1$.
2. **Make a small variation on initial conditions** Select a nearby point separated by d_0 at time i , we picked $d_0 \sim 10^{-6}$ as suggested in the literature.
3. **Advance deviated orbits and compute the new separation** Advance by one iteration $(i+1)$ and calculate the new separation d_1 , here we have taken as an approximation the euclidean distance as

$$d = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2 + (z_a - z_b)^2}$$

where subscripts a and b denote the reference and deviated orbits respectively, complete by evaluating at each time-step the fraction :

$$\frac{\log(d_1/d_0)}{i \times \Delta t}$$

4. **Readjust the deviated orbit** After one iteration of the deviated orbit, move it back to the vicinity of the reference orbit at a distance d_0 again ! This step is also called *re-initialization* or *re-normalization*, it is done for example for x as

$$x_{b,new}^{(i+1)} = x_a^{(i+1)} + d_0 \frac{x_{b,past}^{(i+1)} - x_a^{(i+1)}}{d_1}$$

5. **Repeat steps 3-4 many times** Iterate last two steps as many as needed and that's it !

We have plotted in Figure 4, the behavior of the Lyapunov exponent for different values of ϵ and κ with respect to iterations time-steps. Let's analyze it step by step,

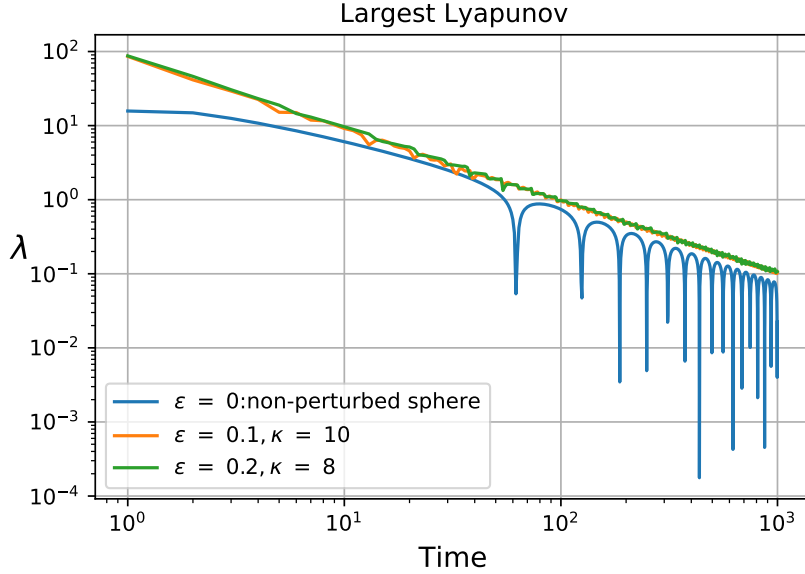


Figure 4. Time dependence of the geometrical Lyapunov exponent for trajectories of 1st, 2nd and 3rd kind. Log-Log scale is used to evidence the 1/t decay.

- For $\epsilon = 0$ (i.e. non-perturbed sphere $\equiv$ trajectories of 1st kind), where the geodesics are arc of circles, the slope of λ is approximately of -1 with pics of irregularities and as it is plotted in log-log, this has to be understood as it evolve as $1/t$, this is significant of non-chaotic system. This is due to the fact that orbits in this situation maintain a constant separation.

- For $\epsilon \neq 0$ (i.e. trajectories of 2^{nd} and 3^{rd} kind), to curve of the Lyapunov exponent conserve it's negative slope but we see clearly that there is no irregularity pics.
- Maybe due to the form of the perturbation [Equation 3.9](#) we have chosen (the simplicity of the system), but we were not able to differentiate between trajectories of 2^{nd} and 3^{rd} kind using the Lyapunov exponent, is it why we proposed another observable presented in below section.

Theses results have also been checked for smaller time-steps $\Delta t = 0.01$ and 0.001 and largest perturbations parameter ϵ , the only difference is for $\epsilon \neq 0$, we observed an horizontal stability at certain point.

5 Jacobi-Levi-Civita equation

Another way of doing but still equivalent is to solve the so-called *Jacobi-Levi-Civita* equation, hopefully for our case and it was shown in [3] that JLC equation [2.2](#) for a two-dimensional manifold simplify enormously and writes,

$$\frac{d^2 Y}{ds^2} + \frac{1}{2} \mathcal{R} Y = 0, \quad (5.1)$$

where $\mathcal{R}$ denotes the scalar curvature of the manifold, the Ricci tensor is given by,

$$R_{jk} = \partial_i \Gamma_{jk}^i - \partial_j \Gamma_{ki}^i + \Gamma_{ip}^i \Gamma_{jk}^p - \Gamma_{jp}^i \Gamma_{ik}^p \quad (5.2)$$

and the Ricci scalar is given by the Trace as,

$$\mathcal{R} = g^{jk} R_{jk} \quad (5.3)$$

Unperturbed sphere

For the sphere, the result is well known,

$$\mathcal{R} = \frac{2}{r}, \quad (5.4)$$

it is simply a constant-curvature AND positive (isotropic) manifolds, for which the JLC equation [5.1](#) simplifies even more for $r = 1$ to,

$$\frac{d^2 Y}{ds^2} + Y = 0. \quad (5.5)$$

Which is simply and Harmonic oscillator equation ! Thus, the norm of the separation vector J does not grow and the system is stable as explained in [section 2](#).

More generally, for a manifold of constant positive curvature, the distance between any pair of nearby geodesics issuing from a neighborhood is oscillating and bounded from above. This is illustrated in [Figure 5](#) by the blue shape.

Perturbed sphere

The Ricci scalar is no longer constant, but is a function of $R(\theta, \phi)$ (the general form have been putted in [A](#)), the parametric instability of [Equation 5.1](#) have been checked for some trajectories of 2^{nd} and 3^{rd} kind and illustrated also in [Figure 5](#).

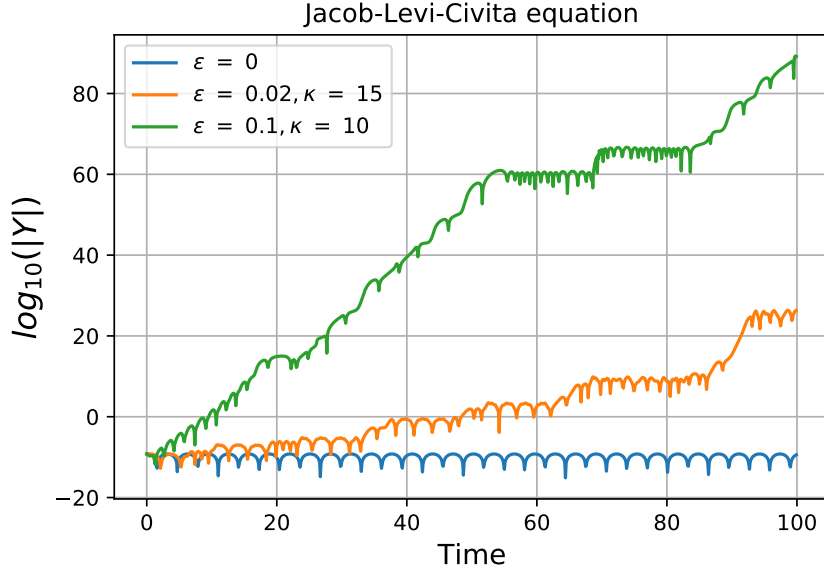


Figure 5. Jacobi-Levi-Civita solutions for trajectories of 1^{st} , 2^{nd} and 3^{rd} kind.

We clearly observe that the solution grows more exponentially with time as the trajectory becomes more chaotic. Thus, the conclusion if this part reads as follows,

chaos is due to the parametric instability of nearby geodesics induced by fluctuations of positive curvature

6 Summary and perspectives

In what follows, we will firstly summarize our previously explained work and give a hint to some future orientations respectively.

Results & conclusions

So in conclusion, here is the most important points to remember,

1. It seems that even if the system lost completely the Integrability memory, it is not always linked to positive Lyapunov exponent.
2. Jacobi-Levi-Civita solution seems more sensitive to chaoticity and thus, solving the Jacobi-Levi-Civita equation may serve as a better probe of chaoticity than the Largest Lyapunov exponent, at least for this model.
3. Riemannian-geometric description of Hamiltonian dynamics opens the doors for new observable such the ones studied here, dynamical systems are far from having given us all their secrets.

Future directions

For sure, our work is not complete and there is still a lot of things that can be done, here is some ideas on what can be performed to refine our results and conclusions,

- Computing Largest Lyapunov exponent based on orbit separation is not the only way of doing, they also exist a more complete way of doing known as *continuous Gram-Schmidt orthonormalization*. It could be interesting to verify our previous results by doing so.
- Even with the available code, it should exist some form of the perturbation for which results can be suitable for more interpretations.
- The model builded here is valid for classical dynamical systems, it could be interesting to study more exotic situations such as the Torus $\mathcal{T}^2$, where from the beginning and without any perturbation, we have access to trajectories of 3^{rd} kind.

Acknowledgements

The author warmly acknowledge Pr. Marco Pettini (Aix-Marseille Univ, Université de Toulon, CNS, Centre de Physique Théorique Marseille) for guiding advises, inspiring mentoring and a huge amount of patience all along this internship.

Appendix A Ricci scalar for perturbed metric

Notice that the Ricci scalar is denoted as $\mathcal{R}$ at the beginning of first line for convenience and that R has to be understood as $R(\theta, \phi)$,

$$\begin{aligned}
\mathcal{R} \times & \left[(R+1) \left(\frac{\partial R^2}{\partial \phi} + \sin^2(\theta) \left(\frac{\partial R^2}{\partial \theta} + R(R+2) \right) + \sin^2(\theta) \right)^2 \right] = \\
& - 2 \sin^2(\theta) R^2 \left(\left(\frac{\partial R}{\partial \phi} \right)^2 + \sin(\theta) \left(\sin(\theta) \left(\left(\frac{\partial R}{\partial \theta} \right)^2 - 3 \right) + \cos(\theta) \frac{\partial R}{\partial \theta} \right) \right) \\
& + \sin^2(\theta) \left(-2 \left(\frac{\partial R}{\partial \theta} \frac{\partial R}{\partial \phi} \right)^2 - \left(2 \frac{\partial R^2}{\partial \theta} + 1 \right) \left(2 \left(\frac{\partial R}{\partial \theta} \right)^2 + \sin(2\theta) \frac{\partial R}{\partial \theta} + \cos(2\theta) - 1 \right) \right) \\
& + \sin^2(\theta) \left(\frac{\partial R}{\partial \theta} \right)^2 \left(2 \left(\frac{\partial R}{\partial \theta} \right)^2 + \sin(2\theta) \frac{\partial R}{\partial \theta} + \cos(2\theta) - 1 \right) \\
& + R \left((1 - 3 \cos(2\theta)) \frac{\partial R^2}{\partial \phi} + 2 \sin^2(\theta) \left(- \left(\frac{\partial R}{\partial \theta} \frac{\partial R}{\partial \phi} \right)^2 + \left(\frac{\partial R}{\partial \theta} \right)^2 \left(\left(\frac{\partial R}{\partial \theta} \right)^2 - 2 \right) \right) \right) \\
& + 2R \sin^3(\theta) \left(2 \sin(\theta) \left(\left(\frac{\partial R}{\partial \theta} \right)^2 - \left(\frac{\partial^2 R}{\partial \theta^2} \right) \right) + \cos(\theta) \frac{\partial R}{\partial \theta} \left(\left(\frac{\partial R}{\partial \theta} \right)^2 - 2 \right) \right) \\
& + \left(4 \sin(\theta) \cos(\theta) \frac{\partial R}{\partial \phi} \frac{\partial R}{\partial \theta} \frac{\partial R}{\partial \phi} \right) - \frac{\partial R^2}{\partial \phi} \left(4 \sin^2(\theta) \left(\frac{\partial R}{\partial \theta} \right)^2 + 4 \sin(2\theta) \frac{\partial R}{\partial \theta} + 3 \cos(2\theta) - 1 \right) \\
& + 3 \sin^2(\theta) + 4 \sin(\theta) \frac{\partial R}{\partial \phi} \frac{\partial R}{\partial \theta} \frac{\partial R}{\partial \phi} \left(2 \sin(\theta) \frac{\partial R}{\partial \theta} + \cos(\theta) \right) + 2 \sin^4(\theta) R^3
\end{aligned}$$

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